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M1 MACRO

Group A

Session 6

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

Model with production

$$1: C_1 + K + S = Y_1$$

$\underbrace{\quad\quad\quad}$ $\underbrace{\quad\quad\quad}$

saving using lending

↳ saving using investment

$$\underline{2} \quad C_2 = Y_2 + (1-\tau_2)S + F(K)$$

Houehold problem

$$\text{Max } \log c_1 + \beta \log c_2$$

$$\text{s.t. } p_1 c_1 + p_2 c_2 + \cancel{pX} = \cancel{P_1 y_1} + \cancel{P_2 y_2} + \cancel{pX + \pi}$$

c_1, c_2, X

$$\mathcal{L} = \log c_1 + \beta \log c_2 + \lambda (P_1 y_1 + P_2 y_2 + pX + \pi - p_1 c_1 - p_2 c_2 - pX)$$

$$\text{FOC } \left\{ \begin{array}{l} /c_1 \quad \frac{1}{c_1} = \lambda p_1 \\ /c_2 \quad \beta /c_2 = \lambda p_2 \end{array} \right\} \Rightarrow \boxed{\frac{c_2}{\beta c_1} = \frac{p_1}{p_2}}$$

$$\left\{ \begin{array}{l} /X \quad \lambda (\underline{p} - p_1) = 0 \end{array} \right\} \Rightarrow \text{only } p = p_1 \text{ can be an equilibrium}$$

$$c_1 = \dots$$

$$c_2 = \dots$$

↙ functions of $\overline{y}, \overline{\pi}, \beta_1, \beta_2$

Remark

Nat. Accounting
closed economy,
no govt

$$(1) \quad Y = C + \underline{I} \quad \text{good market}$$

B> def

$$S = Y - C \quad (2)$$

$$(1) \wedge (2) \Rightarrow I = S$$

2 Firms

Technology of 1 firm:

$\left\{ \begin{array}{l} 1 \text{ u of input} \\ 2 \text{ u of } \text{---} \end{array} \right.$

$\begin{array}{l} 2 \text{ u. of output} \\ 3 \text{ ---} \end{array}$

$\left. \begin{array}{l} 2 \text{ units of inputs} \end{array} \right\} \begin{array}{l} 2 \text{ u in 1 firm} \rightarrow 3 \\ 1 \text{ u in each firm} \rightarrow 4 \end{array}$